

An Experiment to Form a Radio Communication Channel in a Marine Environment

A. K. Tomilin^{a*}, A. F. Lukin^b, and A. N. Gulkov^b

^a National Research Tomsk Polytechnic University, Tomsk, 634050 Russia

^b Far Eastern Federal University, Vladivostok, 690090 Russia

*e-mail: aktomilin@tpu.ru

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Abstract—A field experiment on the transmission of a short-wave modulated radio signal over a distance of 470 m in a marine environment using spherical antennas is described. Special transceiver equipment has been designed based on the results of the generalized electrodynamic theory. It is hypothesized that a high-frequency radio communication channel can be created in a marine environment by means of longitudinal electromagnetic waves.

Keywords: radio communication in a marine environment, generalized electrodynamics, spherical antenna.

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The problems of creating terrestrial wireless radio communication channels were successfully solved more than a hundred years ago, but no analogous system for communication with autonomous underwater objects has appeared so far. This is due to the significant attenuation of high-frequency transverse electromagnetic waves in an electrically conductive medium. The use of ultralow frequencies is restricted by the need to apply large antennas and insufficient data transmission rate [1, 2]. Hydroacoustic communication is often used for the contacts with mobile underwater objects, which also has significant drawbacks [3, 4].

Currently, a generalized electrodynamic theory is being developed that takes into account both vortex and potential electromagnetic processes [5–9]. The potential component of the magnetic field, which is described by the scalar induction function $B^*(\mathbf{r}, t)$, has been found experimentally. The conditions for the appearance of a potential magnetic field and its properties have been determined. A theory of a complex electromagnetic wave has been contrived, which describes it as a sequence of vortex and potential processes. As a result of the vortex processes, transverse electromagnetic waves are generated that are determined by vectors $\mathbf{E}_{\text{rot}}$ and $\mathbf{H}$. Potential processes generate longitudinal electromagnetic waves, which are described using scalar potential of the electric field $\phi(\mathbf{r}, t)$ and scalar function $B^*(\mathbf{r}, t)$. Since the electric potential determines the potential electric field ($\mathbf{E}_{\text{grad}} = -\nabla\phi$), longitudinal electromagnetic waves are usually called “electroscalar waves” (or “scalar-longitudinal waves”) [7–9].

The goal of this work is to experimentally verify the possibility of creating a short-wave radio communication channel for longitudinal waves in marine environment.

To technically implement radio communication in marine environment, it is necessary to create a nonstationary spherically symmetric high-intensity electric field. This is possible due to the phenomenon of irrotational electromagnetic induction [5–9], which is described by the differential equation

$$\nabla \cdot \mathbf{E}_{\text{grad}} = \frac{\rho}{\epsilon_0 \epsilon'} + \frac{\partial B^*}{\partial t}. \quad (1)$$

It follows from Eq. (1) that a potential electric field can be generated not only with the help of electric charges of density ρ , but also due to a nonstationary potential magnetic field, which creates a quasi-charge of density $\epsilon_0 \epsilon' \partial B^* / \partial t$. Monograph [5] describes experiments confirming this phenomenon.

This idea can be realized technically with the help of a Tesla flat spiral coil when a nonstationary current propagates through it [5, 8, 9]. A nonstationary high-density quasi-charge is formed in its center. If the center of the coil is connected to a metal sphere, then potential $\phi(t)$ appears on the sphere, varying with a large amplitude. This forms a strong potential electric field around the sphere. It is nonstationary and has a radial structure.

A method for calculating parameters of the transmitting–receiving equipment was developed, and experimental samples were made on the basis of the theory described in monograph [5] (Fig. 1). A Tesla

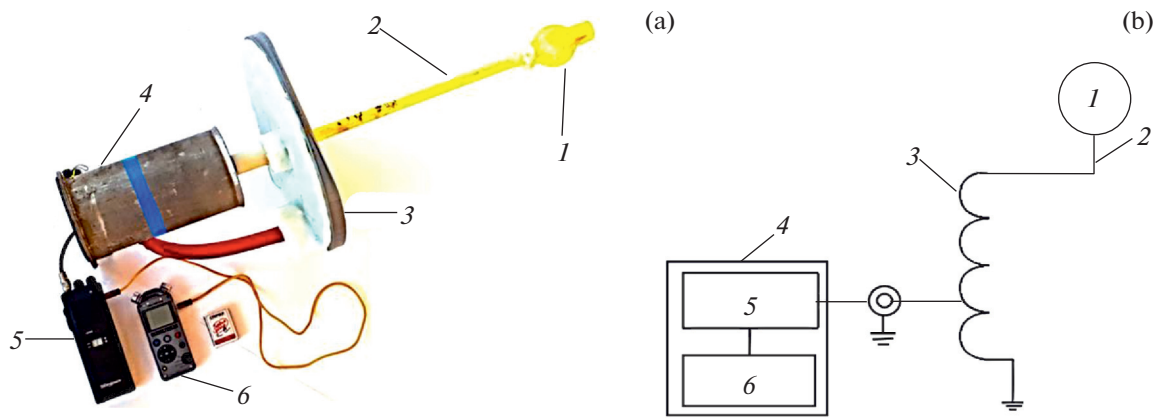


Fig. 1. Experimental sample of (a) the receiving station and (b) its electrical circuit. (1) Copper sphere (antenna), (2) rod, (3) Tesla coil, (4) sealed metal box, (5) radio station, and (6) digital voice recorder.

coil with a diameter of 320 mm has 27 loops of copper wire with a diameter of 2 mm. The central loop of the coil is connected by a metal rod to a copper sphere 60 mm in diameter. A signal from the output of the Shturman-882M radio station at a frequency of 27.4 MHz is sent to the input of the transmitting antenna. The signal amplitude is modulated by the frequency of the “tone call,” the spectrum of which is in the frequency band of about 498–511 Hz. The power of the radio station for radiation is 1.17 W. The effective high-frequency voltage at the output of the transmitter when operating on a real load is 7.66 V. The effective current in the coil is 0.15 A. As a result of the transformation in the Tesla coil, the amplitude of the high-frequency voltage on the spherical antenna is approximately 45 V. This corresponds to a nonstationary charge varying with an amplitude of 12.5×10^{-9} C, which is generated on a copper-emitting sphere in an aqueous medium.

The radio transmitter, together with the power supply batteries, is located in a sealed box rigidly connected to the antenna. All antenna elements are insulated with a dielectric layer from contact with seawater. The receiving station has a similar design. The modulation signal from the headphone jack of the radio in the receiving station is sent to the input of a Olympus LS-12 voice recorder located in a sealed box.

A series of experiments were carried out with a gradual increase in the signal transmission range. The paper presents data on tests that were carried out on October 27, 2020, in Novik Bay (Vladivostok, Russky Island). The receiving station was installed on a moored buoy at a depth of 6 m. The sea depth in this place is 13 m. The conductivity of seawater in the test site was 4.77 S/m. The water temperature was +8°C. Both the receiving and transmitting stations were lowered into the water on nylon ropes without metal parts. This ensured the complete electromagnetic autonomy of both modules in a submerged state. The transmitting station was completely immersed in the water and

was located at a depth of 4 m from the sea surface. From the 9th to the 40th minute, the transmitting station was evenly towed by the boat in the direction from the receiving module. The distance between the stations was measured using a GPS receiver.

The signal recorded on the voice recorder was processed using spectral analysis in a 30-s window with a 7.5-s window step and a Gaussian weight window. The average power of the “ringing tone” signal for the modulated signal (MS) of the transmitting radio station in the frequency band of 498–511 Hz, referred to the frequency band of 1 Hz, has been calculated. For comparison, in the frequency band of 515–520 Hz, where there were no frequency components of the “ringing tone” signal, the average background noise power (NP) was calculated, which was also referred to the frequency band of 1 Hz. Spectral analysis of a signal from a calibrated and verified sound generator recorded on a voice recorder made it possible to recalculate the MS and NP levels in dB relative to a power level of 1 μ W, reduced to a frequency band of 1 Hz. The results depending on the time of the experiment are shown in Fig. 2.

The receiving module was in seawater from the 9th to the 43rd minute of the recording. The distance between the transmitting and receiving modules varied from 20 m on the 9th minute of recording to a maximum of 470.7 m on the 40th minute. The average MS level during the towing was maintained in the range from –87 to –88 dB as a result of the automatic gain control in the receiving radio station. The average NP level during the same time varied from –95 to –93 dB. Thus, the level of the MS excess over the NP level recorded in the experiment was 5–8 dB relative to 1 μ W in the 1-Hz frequency band.

From the zero minute to the 9th minute of recording and from the 43rd to 46th minutes of recording, the transmitting module remained in seawater, and the receiving module was on board the boat. During these periods of time, the MS level

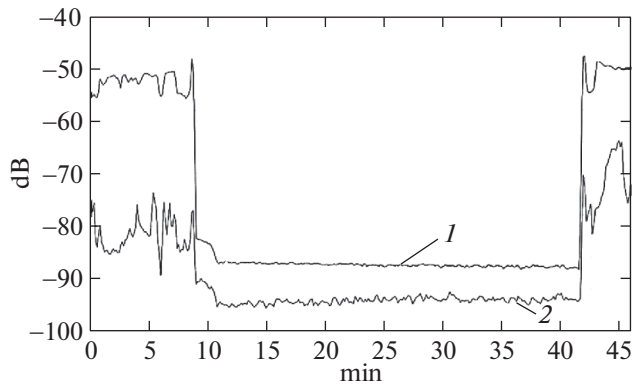


Fig. 2. Mean power of (1) the received signal and (2) background noise relative to the power of 1 μ W referred to the frequency band of 1 Hz.

varied from -48 to -55 dB. At the same time, the NP level varied from -74 to -89 dB before the beginning of towing and from -64 to -81 dB after its end. Thus, the fact of reliable signal transmission from water to air was recorded when the radio signal propagated over a distance of 4 m in water and about 3 m in air, with an excess of the MS level over the NP level by 20 dB or more. In other experiments, the transmission of a radio signal from the air to the sea was also recorded.

The calculated attenuation of the radio signal power level for transversal waves at a distance of 1 m in marine environment with a conductivity of 4.77 S/m at a frequency of 27.4 MHz was about 195 dB/m. At a distance of 470 m, the total attenuation should be about 91 650 dB, which is many times higher than the noise immunity margin of the radio stations used. However, the underwater radio communication channel actually works at this distance. Therefore, it can be assumed that the radio signal is transmitted in seawater using longitudinal (electroscalar) waves.

Thus, the possibility of creating a high-frequency radio communication channel in the marine environment has been proved experimentally under natural conditions. Presumably, this happens using electroscalar waves. The effect of transmission of a high-frequency radio signal across the interface between two media—seawater and air—has been found.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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